

ADASTT: Adaptive Speech-to-Text Algorithm Selection via Meta-Learning

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Abstract—We introduce ADASTT, an adaptive meta-learning framework that selects, in real time, the most suitable speech-to-text (STT) model for each incoming audio input. Factors such as background noise, speaker accent, speaking rate, and recording quality cause different STT models to perform better on different inputs. Exhaustively running every model is computationally expensive and inefficient, whereas committing to a single model degrades accuracy. ADASTT addresses this trade-off by computing a lightweight feature vector for each audio input capturing noise level, speaker traits, temporal dynamics, and signal quality and passing it to a probabilistic meta-learner. The meta-learner assigns a probability distribution over candidate STT models and samples a single model to transcribe the audio input. Experiments show that ADASTT consistently reduces the Word Error Rate (WER) across heterogeneous conditions. On the AMI Meeting corpus, which exhibits high acoustic variability, ADASTT achieves a relative WER reduction of 10.6% over traditional fusion methods while being $2.7\times$ faster than standard ROVER fusion. End-to-end ADASTT is approximately 26.5% faster on average than Whisper Large v3 using much smaller S2T algorithms. The framework is learner-agnostic and meta learner or base STT models can be used. We also present boosting-based and neural learners and provide proofs of convergence to a first-order stationary point under mild regularity conditions.

Index Terms—Automatic speech recognition (ASR), convergence analysis, meta-learning, model selection, speech-to-text (STT).

I. INTRODUCTION

A. Background

REAL time STT transcription is an important component for various modern applications ranging from voice assistants and automatic captioning systems to critical domains such as healthcare [1], [2]. The efficiency of STT systems has significantly improved due to recent advances in deep learning approaches, yet these systems face major difficulties when

operating under realistic, heterogeneous acoustic conditions, which include speaker accent variations, background noise, reverberations, and speech rate variations [3]. The performance of traditional STT models, often trained and optimized on clean datasets, deteriorates significantly when they encounter real-world acoustic variations [3], [4]. Consequently, building robust STT systems capable of consistently high accuracy across diverse acoustic scenarios remains a crucial and unresolved challenge in the audio signal processing and machine learning communities, even with recent efforts to improve real-world robustness through large-scale weak supervision and unified pretraining objectives [5], [6], [7]. Here, we introduce ADASTT, a meta-learning-based framework that adaptively selects the most suitable STT model for each audio input in real time. By efficiently extracting discriminative audio features and selecting the most appropriate STT algorithm based on a learned probabilistic selection policy, ADASTT bolsters system robustness by avoiding major model failures through intelligent routing. This approach ensures high transcription reliability across diverse acoustic conditions with minimal computational overhead compared to running a single algorithm.

ADASTT is structured as a two-phase framework: an offline training phase and an online inference phase. We emphasize that this structure can be readily extended to full online case by a sliding window approach, e.g., by updating the meta learner with new audio data, for non stationary environments. In the offline phase, a meta-learner is trained using a reference dataset comprising diverse acoustic conditions, with a custom training objective optimizing the transcription accuracy. In the subsequent online inference phase, the trained meta-learner quickly identifies the best-suited transcription algorithm for each input based on discriminative audio-derived features. Unlike conventional methods that rely on static model selection or computationally intensive ensemble combinations [8], ADASTT applies a data-driven, adaptive selection process. This capability allows to achieve superior transcription performance across a wide range of realistic acoustic scenarios, and we further support this with convergence derivations that characterize the selection policy's suitability under non-stationary conditions.

B. Related Work

Automatic speech recognition (ASR) has advanced rapidly over the past decade due to deep learning and the availability of large-scale datasets. End-to-end neural models now achieve low error rates on clean benchmarks such as LibriSpeech [9].

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The implementation of ADASTT is publicly available at <https://github.com/farukclkk/ADASTT>.

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However, real-world speech exhibits substantial variability in accent, background noise, reverberation, and speaking style, under which recognition performance can degrade significantly when test conditions differ from training data [3]. Even state-of-the-art (SOTA) models such as Wav2Vec 2.0, WavLM, and Whisper, while robust in clean settings, show performance degradation on noisy or out-of-distribution audio [5], [6], [7], [10], [11]. While data augmentation and multi-style training improve robustness [4], [12], they do not adapt dynamically to the characteristics of individual inputs. Our work addresses this limitation by introducing a meta-learned controller that selects the most suitable pretrained STT model on a per-input basis.

At the architectural level, robustness has been pursued through increasingly expressive neural models. Early work demonstrated the benefits of convolutional and hybrid architectures for noisy speech [13], [14], while more recent approaches integrate self-attention and convolutional modules. The Conformer architecture jointly models local and global context and has become a strong backbone for robust ASR [2]. Streaming variants such as the Transformer-Transducer further enable low-latency recognition under challenging conditions [15]. Despite their success, these approaches deploy a fixed architecture at inference time and cannot adapt model choice to input-specific conditions. Our method is complementary: rather than designing a single universally robust model, we learn a lightweight policy that dynamically selects among pretrained STT engines.

Recent work has also explored parameter-efficient adaptation for robustness and low-resource ASR. Adapter-based methods enable cross-lingual and low-resource transfer by inserting lightweight modules into pretrained models [16]. Meta-learning has been applied on top of such architectures to rapidly adapt self-supervised speech models via meta-adapters, improving performance under limited supervision [17]. More recently, Meta-Whisper introduces speech-based meta in-context learning for low-resource ASR, enabling rapid adaptation within a single model without full fine-tuning [18]. While effective, these approaches focus on adapting or reparameterizing a single base model. In contrast, our framework operates at the system level, selecting among multiple pretrained STT engines without modifying their internal architectures.

Recognizing that no single STT model consistently outperforms others across all acoustic scenarios [3], prior work has explored ensemble learning, algorithm selection, and meta-learning. Classical ensemble methods such as bagging, boosting [19], [20], and voting-based schemes like ROVER [8] combine multiple models using static rules. More recent approaches perform dynamic selection based on input features. The META-DES framework estimates the competence of each base learner per instance [21], while Arunkumar et al. [22] use similarity measures over self-supervised speech embeddings to derive adaptive fusion weights. Mixture-of-experts (MoE) routing has also been explored in ASR, activating subsets of experts to improve accuracy with limited compute [23], [24]. Although effective, these methods often require architectural modifications or incur additional training and inference overhead, limiting their practicality for real-time selection among pretrained STT systems.

Meta-learning has further been investigated for rapid adaptation in ASR, particularly for speaker and accent adaptation. Methods such as MAML [25] and Prototypical Networks [26] enable fast learning from limited data and have been applied to speaker personalization and accent adaptation [27], [28]. However, these approaches focus on fine-tuning a single model rather than selecting among multiple experts at run time. The direct application of meta-learning to real-time, per-input STT model selection remains relatively unexplored. Our work fills this gap by introducing a lightweight, feature-conditioned meta-learner that performs dynamic algorithm selection with negligible inference overhead, enabling robust transcription across heterogeneous, real-world acoustic conditions.

C. Contributions

Our contributions are summarized as follows:

- We introduce an approach, which chooses the most suitable STT algorithm for each audio by using a meta learner to achieve the oracle loss rate with minimal computational overhead over a single STT algorithm in the ensemble.
- Our approach is loss agnostic, i.e., our algorithm can optimize any performance measure including WER, CER, and similar [3].
- Our method selects the most suitable STT model for each audio input using a lightweight meta-learner trained on easily extractable acoustic and semantic features, and is feature-vector-agnostic: any set of discriminative audio features can be used without requiring changes to other components of the algorithm, while introducing only minimal computational overhead beyond a single model run.
- The framework is meta-learner agnostic, allowing for flexible integration of various learning algorithms (e.g., decision trees, neural networks, or gradient-boosted models) without requiring any change to the core selection mechanism.

D. Organization

The remainder of the paper is organized as follows. Section II formulates the problem under consideration. Section III introduces the ADA-STT framework and its main components. Section III-C describes the two instantiated meta-learners and details their training objectives and optimization procedures. Section IV presents convergence guarantees to first-order stationary points for the proposed learners. Section V reports the experimental setup, results, and analysis, while Section V-F provides an ablation study of the employed feature sets. Finally, Section VI concludes the paper. Detailed convergence proofs are provided in Appendices A and B.

II. PROBLEM DESCRIPTION

We study real-time STT transcription for a large volume of streaming audio data denoted as $\{a_n\}_{n=1}^N$, where N is significantly large. As for most real life applications, we assume that characteristics and features of each audio clip such as background noise levels, speaker accent, speaking rate, recording

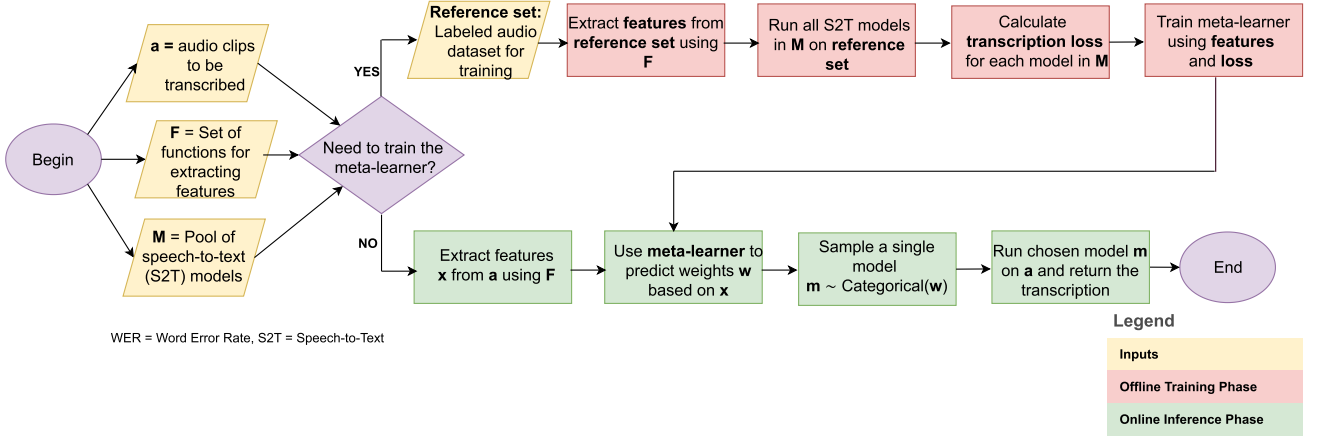


Fig. 1. The overall architecture of ADASTT algorithm. The training phase, where a meta-learner is trained using labeled training data, is shown in red, while the inference phase, where features guide the selection of a single STT algorithm using the trained meta-learner, is shown in green.

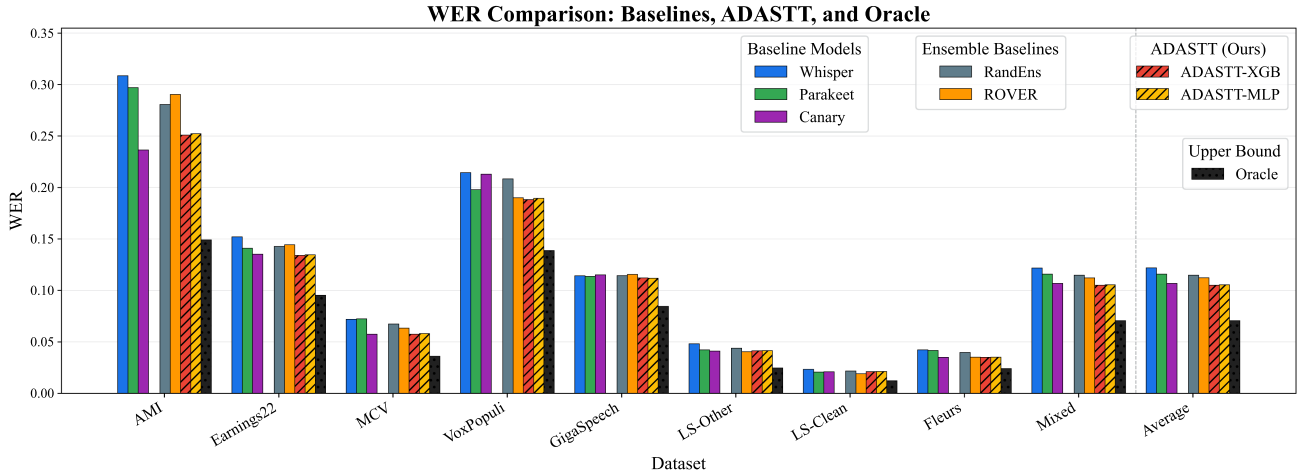


Fig. 2. WER Comparison Across Various Datasets.

quality, and acoustic clarity significantly vary. Naturally, due to this diversity, different state of the art STT algorithms show varying levels of transcription performance depending on these audio specific features [1], [3], and it is unknown in advance which algorithm will perform the best for a given audio input. Applying all available STT algorithms to every incoming audio clip is computationally expensive and inefficient. Also, for each clip, selecting the best STT algorithm from a finite set of K distinct STT algorithms with little computational overhead is challenging. Here, we introduce a highly effective and computationally efficient approach which performs this selection.

Let $T_{n,m}$ represent the transcription produced by the m -th STT algorithm for an audio clip a_n , i.e., $T_{n,m} = M_m(a_n)$, where $\mathcal{M} = \{M_1, M_2, \dots, M_K\}$ denotes the set of all STT algorithms, and $M_m \in \mathcal{M}$ is the m -th algorithm. Each transcription $T_{n,m}$ is evaluated through a fitness measure, defined using standard metrics such as the WER, or analogous metrics suitable for transcription accuracy assessment [3]. Given any of these measures, we denote the transcription error, or the loss incurred when using the STT algorithm m to transcribe the audio clip a_n , as $\ell(T_{n,m}, R_n)$, where R_n is the ground-truth transcription associated with the audio clip a_n , and ℓ , for example, can be any

error measure. Our goal is to select the most suitable algorithm for each incoming a_n in an effective way in order to minimize the total transcription error:

$$\sum_{n=1}^N \ell(T_{n,m}, R_n)$$

where M_m is the selected algorithm for the n th audio clip producing $T_{n,m}$. We next introduce an algorithm that uses a meta-learner to compute a probability distribution over the STT algorithms based on features extracted from the input audio a_n . The algorithm selects a single STT algorithm from this computed probability distribution to obtain the final transcription in an effective manner.

III. ADASTT: ADAPTIVE STT ALGORITHM SELECTION VIA META- LEARNING

A. Algorithm Overview

ADASTT consists of an offline training phase and an online inference phase as shown in Fig. 1. In the offline phase, the meta-learner learns how to assign probability weights to STT

Algorithm 1: Online STT Algorithm Selection via ADASTT.

Inputs:

- a_n : Unlabeled input audio clip
- f_θ : Trained meta-learner based selection algorithm
- $\mathbb{F}$: Set of audio feature extraction functions
- $\mathbb{M} = \{M_1, M_2, \dots, M_K\}$: Pool of K STT algorithms

Output:

- Final transcription $\hat{T}_n$ for audio a_n

Inference Steps:

- 1: Extract feature vector: $\mathbf{x}_n \leftarrow \mathbb{F}(a_n)$
 - 2: Predict selection logits: $\mathbf{z}_n \leftarrow f_\theta(\mathbf{x}_n)$ ▷ See Section III-C
 - 3: Normalize $\mathbf{z}_n$ via softmax to obtain weights $\mathbf{w}_n$
 - 4: Sample algorithm $\hat{m} \sim \text{Categorical}(\mathbf{w}_n)$ ▷ See Section III-D
 - 5: Return final transcription: $\hat{T}_n \leftarrow M_{\hat{m}}(a_n)$
-

algorithms based on features extracted from labeled audio clips. In the online phase, given an input audio a_n , we extract a feature vector $\mathbf{x}_n$, and use the trained meta-learner $f_\theta(\mathbf{x}_n)$ to produce a weight vector $\mathbf{w}_n \in [0, 1]^K$ over the candidate STT algorithms $\{M_1, \dots, M_K\}$, where θ denotes the learner parameters for the meta-learner. We then select a single algorithm M_m using this probability distribution and return its transcription $T_{n,m}$. The online inference steps are summarized in Algorithm 1, and the full pipeline is illustrated in Fig. 1.

B. Feature Set

We represent each audio clip a_n with a feature vector $\mathbf{x}_n$ that captures acoustic, prosodic, semantic, and temporal features relevant to STT performance [29]. All features are represented at the utterance level: frame-level descriptors are aggregated using standard temporal pooling statistics (e.g., mean and variance), while signal-level features are computed directly over the full audio segment.

a) Spectral-Phonetic Features (MFCC): Mel-frequency cepstral coefficients (MFCCs), first proposed by Davis and Mermelestein [30], represent the spectral envelope of speech signals and effectively distinguish phonetic content at the phoneme level. Following long-established ASR practice [29], [30], we extract the standard set of 13 static MFCCs (coefficients 0–12) along with their first- and second-order derivatives. Lower-order coefficients (0–5) capture the broad spectral envelope, while higher-order terms (6–12) encode finer formant details necessary for discriminating fricatives, nasals, and high-frequency consonants. Δ and $\Delta\Delta$ terms represent temporal dynamics, enhancing robustness to coarticulation and speaker variation [29]. Including all 13 coefficients has repeatedly been shown to reduce phoneme error rates [31]. Frame-level MFCC features are summarized into utterance-level statistics before being passed to the meta-learner.

b) Prosodic Features (Pitch and VAD): Prosodic features such as pitch (mean, range) and VAD-based silence ratios encode speaking style, rhythm, and ambient noise levels [32], [33], [34].

These features are included to help the meta-learner identify STT algorithms that are sensitive to prosodic variation, which often leads to higher transcription errors. While the original Whisper and Wav2Vec2 papers do not report performance broken down by pitch or prosodic characteristics [5], [10], recent benchmarks show that Whisper outperforms other STT models on emotionally expressive datasets [35].

c) Neural Speech Embeddings: We extract high-level contextual embeddings from pretrained STT encoders such as Wav2Vec2 [10], and HuBERT [36]. These speech embeddings encode complex representations, which capture speaker identity, environmental contexts, and semantic structures beyond traditional acoustic parameters. The resulting frame-level embeddings are temporally pooled to form a fixed-dimensional utterance-level representation. We emphasize that this embedding extraction step is computationally efficient, i.e., compared to the full decoding, as it requires only a single forward pass through the encoder without full decoding or language modeling. Hence, these features can be extracted with minimal computational cost compared to running a full transcription.

d) Signal and Temporal Complexity Features: To quantify signal clarity and spectral structure, we extract signal-to-noise ratio (SNR), zero-crossing rate (ZCR), and spectral entropy. These features are shown to encode noise intensity, perceptual harshness, and spectral complexity, respectively [34], [37]. Additionally, we extract global timing descriptors such as speech segment duration, speech rate (voiced frames/s), and pause ratio. These factors are known to affect ASR performance in rapid, noisy, or disfluent speech [4], [15], [38]. These quantities are computed directly over each full audio segment without additional preprocessing.

The above feature set is chosen as a reasonable and representative baseline for heterogeneous speech conditions, balancing low-level acoustic cues, high-level contextual information, and computational efficiency. Hence, the resulting feature vector describing a_n , $\mathbf{x}_n$ contains low-level acoustic features, prosodic indicators, contextual embeddings, and signal complexity metrics for robust algorithm selection.

Remark 1: Our framework is flexible and feature-agnostic: The proposed framework does not assume or require any specific feature family. The meta-learner operates on a generic feature vector and can incorporate any set of discriminative features such as acoustic, linguistic, prosodic, model-intrinsic, or domain-specific features that correlate with relative STT performance. If alternative features provide better separation between candidate algorithms for a given application, they can be substituted or augmented without modifying the training procedure, loss function, or selection mechanism.

C. Meta-Learner

Given an input feature vector $\mathbf{x}_n \in \mathbb{R}^d$ for the n -th audio sample where d denotes the total number of extracted features, the meta-learner is defined as a function

$$f_\theta(\mathbf{x}_n) = \mathbf{z}_n \in \mathbb{R}^K,$$

where θ are the trainable parameters and K is the number of candidate STT algorithms. A softmax operation is applied to the logits $\mathbf{z}_n$ to obtain the selection probabilities for candidate STT algorithms, which will then be used to select one of the algorithms as shown in Fig. 1. The meta-learner can be instantiated with various trainable algorithms. Here we detail the training of two algorithms, XGBoost, and a feedforward neural network since these algorithms show remarkable performance in widely known machine learning competitions [39], [40].

a) XGBoost-based Learner: XGBoost is a decision-tree-based gradient boosting algorithm, which constructs a set of decision trees to minimize a loss function iteratively [41]. We choose XGBoost for its strong performance on structured input features and built-in regularization mechanisms that reduce overfitting [41]. At inference time, the trained XGBoost model defines a function $f_\theta: \mathbb{R}^d \rightarrow \mathbb{R}^K$ that maps input feature vectors $\mathbf{x}_n \in \mathbb{R}^d$ to unnormalized logits $\mathbf{z}_n = f_\theta(\mathbf{x}_n) = (z_{n,1}, \dots, z_{n,K})$, where each $z_{n,k}$ corresponds to an STT algorithm. The logits $\mathbf{z}_n$ are normalized via a softmax function to obtain a probability distribution over the candidate STT algorithms, from which one algorithm is sampled at inference time (see Fig. 1 and Algorithm 1):

$$w_{n,m} = \frac{e^{z_{n,m}}}{\sum_{j=1}^K e^{z_{n,j}}}, \quad \text{where} \quad \sum_{m=1}^K w_{n,m} = 1. \quad (1)$$

We train XGBoost as follows. Let $\mathbb{M}$ denote the pool of K STT algorithms, and $\mathbb{F}$ the set of feature extraction functions, as shown in Fig. 1. For each training audio clip a_n , we run all STT algorithms in $\mathbb{M}$ to obtain transcriptions $T_{n,m}$, and extract feature vectors $\mathbf{x}_n$ using $\mathbb{F}$. The meta-learner receives $\mathbf{x}_n$ as input and produces unnormalized logits, which are transformed via softmax into a probability vector $\mathbf{w}_n = (w_{n,1}, \dots, w_{n,K})$ over the candidate algorithms. Given the weight vector $\mathbf{w}_n$, we define the expected loss for a single audio clip a_n as

$$\mathcal{L}_n = \sum_{m=1}^K w_{n,m} \text{WER}(T_{n,m}, R_n) \quad (2)$$

where R_n is the reference transcription for audio a_n . The WER between a hypothesis transcription $T_{n,m}$ and the reference R_n is defined as $\text{WER}(T_{n,m}, R_n) = \frac{S+I+D}{|R_n|}$, where S , I , and D denote the numbers of substitutions, insertions, and deletions obtained from a standard Levenshtein alignment [3].

Remark 2: In this work, we choose to minimize the WER due to its simplicity, widespread adoption in the speech recognition community, and its direct interpretability as a measure of transcription quality [1], [3]. Any other transcription accuracy or error metric can also be used in our generic algorithm.

We optimize the XGBoost algorithm parameterized by θ to minimize the total expected transcription error across the training set. The training objective is defined as:

$$\min_{\theta} L = \frac{1}{N} \sum_{n=1}^N \mathcal{L}_n = \frac{1}{N} \sum_{n=1}^N \sum_{m=1}^K w_{n,m} \text{WER}(T_{n,m}, R_n). \quad (3)$$

XGBoost optimizes predictions through second-order Taylor expansions of the loss function, requiring both the first-order (gradient) and second-order (Hessian) derivatives. At each boosting iteration, the algorithm fits a regression tree to the negative gradients, then computes an analytic weight for every leaf that minimizes a regularized quadratic approximation of the loss. To train XGBoost in our setting, we therefore compute the sample-wise gradients and Hessians of $\mathcal{L}_n$ with respect to the logits $z_{n,m}$. Let $g_{n,m}$ be the gradient, and $h_{n,m}$ be the Hessian:

$$\begin{aligned} g_{n,m} &= \frac{\partial \mathcal{L}_n}{\partial z_{n,m}} \\ &= \sum_{k=1}^K \text{WER}(T_{n,k}, R_n) w_{n,k} (\delta_{km} - w_{n,m}) \\ &= w_{n,m} (\text{WER}(T_{n,m}, R_n) - \mathcal{L}_n). \end{aligned} \quad (4)$$

In order to reduce computational cost and ensure compatibility with second-order tree splitting, we approximate the full Hessian matrix by its diagonal, consistent with the training strategy described in XGBoost by Chen and Guestrin [41]:

$$\begin{aligned} h_{n,m} &= \frac{\partial^2 \mathcal{L}_n}{\partial z_{n,m}^2} = \frac{\partial}{\partial z_{n,m}} [w_{n,m} (\text{WER}(T_{n,m}, R_n) - \mathcal{L}_n)] \\ &= \frac{\partial w_{n,m}}{\partial z_{n,m}} (\text{WER}(T_{n,m}, R_n) - \mathcal{L}_n) + w_{n,m} \frac{\partial}{\partial z_{n,m}} (-\mathcal{L}_n) \\ &= w_{n,m} (1 - w_{n,m}) (\text{WER}(T_{n,m}, R_n) - \mathcal{L}_n) - w_{n,m} \frac{\partial \mathcal{L}_n}{\partial z_{n,m}} \\ &= w_{n,m} (1 - 2w_{n,m}) (\text{WER}(T_{n,m}, R_n) - \mathcal{L}_n). \end{aligned} \quad (5)$$

Remark 3: In practice, to satisfy XGBoost's requirement for positive second-order gain estimates during tree building [41], we clip the Hessian values to a small positive constant when $h_{n,m} < 0$. This stabilizes training without significantly impacting convergence. Note that this may happen since we use only the diagonal entries to reduce computational complexity.

These expressions are used to define a custom objective function in the XGBoost training loop, as summarized in Algorithm 2.

b) Neural Network-based Learner: A neural network is a parameterized function composed of layers of interconnected units (neurons), where each layer applies an affine transformation followed by a non-linear activation. This architecture allows neural networks to approximate complex input-output mappings and has been widely adopted in supervised learning tasks [42]. A feedforward neural network, also known as a multilayer perceptron (MLP), is the simplest class of neural networks in which information flows strictly from input to output without recurrence or temporal feedback.

As an alternative to tree-based algorithms, we implement the meta-learner as a feedforward neural network with GELU activations [43]. We train the MLP-based meta-learner analogously to the XGBoost model, as shown in Fig. 1. The key difference is that optimization is performed via stochastic gradient descent instead of second-order boosting. The MLP is optimized via gradient-based learning using the reparameterized expected

Algorithm 2: Offline Training Procedure for Meta-Learner with WER-based Objective.

Inputs:

- $\{(a_1, R_1), \dots, (a_N, R_N)\}$: Labeled dataset of N audio clips and corresponding reference transcriptions.
- $\mathbb{F}$: Feature extraction functions.
- $\mathbb{M} = \{M_1, \dots, M_K\}$: Pool of K candidate STT algorithms.
- Meta-learner architecture (XGBoost or MLP).

Output:Trained meta-learner f_θ **Training Steps**

- 0: **Split** dataset into training and validation sets: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$.
 - 1: **for** $n \leftarrow 1$ **to** $|\mathcal{D}_{\text{train}}|$ **do**
 - 2: Extract features: $\mathbf{x}_n \leftarrow \mathbb{F}(a_n)$
 - 3: **for** each algorithm $m \in \mathbb{M}$ **do**
 - 4: Transcribe: $T_{n,m} \leftarrow M_m(a_n)$
 - 5: Compute: $\text{WER}_{n,m} \leftarrow \text{WER}(T_{n,m}, R_n)$
 - 6: **end for**
 - 7: **end for**
 - 8: Define weights $\mathbf{w}_{n,m}$, instance loss $\mathcal{L}_n$, and training objective $\mathcal{L}$ as in (1), (2), and (3).
 - 9: **if** meta-learner is XGBoost **then**
 - 10: Construct feature matrix $\mathbf{X}$ and WER matrix $\mathbf{W}$.
 - 11: Compute gradients and Hessians as in (4) and (5).
 - 12: Train f_θ using XGBoost with the custom objective.
 - 13: **else**
 - 14: Train f_θ using backpropagation and a gradient-based optimizer (e.g., Adam).
 - 15: **end if**
 - 16: Apply early stopping and tune hyperparameters using validation loss on $\mathcal{D}_{\text{val}}$.
-

WER loss. The WER values $\text{WER}(T_{n,m}, R_n)$ are treated as fixed constants during training, allowing end-to-end optimization over θ without requiring the token-level loss proxy. This enables the MLP to directly learn a distribution that minimizes transcription error. The training steps are summarized in Algorithm 2.

Remark 4: Our framework is flexible: any differentiable supervised learning algorithm such as transformers [44], or other neural architectures can also be used as the meta-learner without modifying the overall pipeline.

D. Sampling Strategy

To complete the inference pipeline, we use a one-shot adaptive sampling strategy to select a single STT algorithm based on the predicted weights. The following pseudocode, Algorithm 3, summarizes this selection procedure. This procedure favours models that are predicted to have better transcription accuracy for a given audio input. However, rather than always choosing the single best predicted model, our method preserves a level of randomness in the selection process. This approach encourages exploration, allowing the algorithm to occasionally select other candidate models. Such exploration ensures robustness by preventing the algorithm from overly relying on predictions

Algorithm 3: One-Shot Sampling from Algorithm Selection Weights.

Input:

- a_n : Input audio clip
- $\mathbf{w}_n = (w_{n,1}, w_{n,2}, \dots, w_{n,K})$: A probability distribution over K STT algorithms for input a_n , satisfying $\sum_{m=1}^K w_{n,m} = 1$
- $\mathbb{M} = \{M_1, M_2, \dots, M_K\}$: Pool of K STT algorithms

Output: Final transcription $\hat{T}_n$ for audio a_n

- 1) Sample a random number: $U \sim \mathcal{U}(0, 1)$
 - 2) Initialize $c_{n,0} \leftarrow 0$ and compute cumulative weights:
 $c_{n,m} = c_{n,m-1} + w_{n,m}$ for $m = 1, \dots, K$
 - 3) Select index: $\hat{m} = \min\{m \mid U \leq c_{n,m}\}$
 - 4) Run selected algorithm on input:
 $\hat{T}_n \leftarrow T_{n,\hat{m}} = M_{\hat{m}}(a_n)$
-

that might occasionally be incorrect, and allows continuous refinement of model selection over time.

E. Fully Online Extension Via Sliding Window Adaptation

In addition to the offline, batch training setting, ADASTT can be extended to fully online and streaming scenarios using a sliding window mechanism. Incoming audio is processed sequentially in short segments, allowing the system to adapt its model selection strategy as new data arrives. This is particularly suitable for real-time applications such as live transcription or interactive speech-based systems.

In the online setting, the meta-learner is updated periodically using information from the most recent window of audio segments. Since ground-truth transcriptions are typically unavailable at inference time, updates rely on proxy performance signals (e.g., internal confidence scores or average log-likelihoods) produced by the selected STT models. The window size controls the trade-off between adaptation speed and stability and can be adjusted based on computational and memory constraints.

For boosting-based meta-learners, online adaptation corresponds to appending a small number of additional boosting steps using the newly collected data, while neural meta-learners perform a limited number of incremental gradient updates. The overall procedure preserves the real-time nature of the system and introduces negligible additional overhead. The resulting online selection process is summarized in Algorithm 4.

IV. MATHEMATICAL PROOF OF CONVERGENCE

We provide convergence guarantees for the two meta-learners presented in Section III-C. Specifically, we show that under mild regularity conditions, both training procedures converge to a first-order stationary point of the training loss function. That is, the norm of the gradient of the loss approaches zero, which guarantees that the models will reach a stable point. This is important because it ensures that the training of the meta-learners is well-behaved and will not diverge or oscillate indefinitely. Although we use a loss based on the WER in experiments,

Algorithm 4: Online ADASTT with Sliding Window Adaptation.

Inputs:

- Stream of audio clips $\{a_n\}_{n \geq 1}$
- Meta-learner f_θ
- Feature extractor $\mathbb{F}$
- STT algorithm pool $\mathbb{M} = \{M_1, \dots, M_K\}$
- Window size W

Initialization:

- Initialize empty buffer $\mathcal{B} \leftarrow \emptyset$

Online Procedure:

- 1: **for** each incoming clip a_t **do**
 - 2: Extract features: $\mathbf{x}_t \leftarrow \mathbb{F}(a_t)$
 - 3: Predict logits: $\mathbf{z}_t \leftarrow f_\theta(\mathbf{x}_t)$
 - 4: Compute selection probabilities: $\mathbf{w}_t \leftarrow \text{softmax}(\mathbf{z}_t)$
 - 5: Sample model: $\hat{m}_t \sim \text{Categorical}(\mathbf{w}_t)$
 - 6: Transcribe: $\hat{T}_t \leftarrow M_{\hat{m}_t}(a_t)$
 - 7: Compute proxy reward r_t (e.g., confidence or log-probability)
 - 8: Append $(\mathbf{x}_t, \hat{m}_t, r_t)$ to buffer $\mathcal{B}$
 - 9: **if** $|\mathcal{B}| = W$ **then**
 - 10: Update meta-learner f_θ using $\mathcal{B}$
 - 11: Clear buffer $\mathcal{B} \leftarrow \emptyset$
 - 12: **end if**
 - 13: **end for**
-

the proofs hold for any smooth loss with bounded curvature, consistent with the framework's loss-agnostic design.

Remark 5: The convergence guarantees presented here apply to the offline training of the meta-learner; the fully online sliding-window extension in Section III-E is intended for practical adaptation and does not rely on these guarantees.

A. XGBoost Meta-Learner

For the XGBoost meta-learner, we prove convergence under our custom WER-based objective by extending standard second-order boosting theory. Assuming bounded loss values, Lipschitz-continuous gradients, and a weak-learner edge condition, we obtain a monotone descent property. Let $L(z^{(t)})$ denote the training loss evaluated at the current logits $z^{(t)}$.

Under these assumptions, and for a sufficiently small step size η , the training dynamics admit the following descent inequality:

$$L_{t+1} \leq L_t - \gamma \|\nabla L(z^{(t)})\|^2, \quad (6)$$

where $\gamma > 0$ is a constant that depends on the learning rate, the weak-learner edge, and the smoothness of the loss function. Summing over iterations yields an $\mathcal{O}(1/T)$ convergence rate to a first-order stationary point. Full derivations and the explicit form of γ are provided in Appendix A.

B. Neural MLP Meta-Learner

For the neural meta-learner, we rely on standard non-convex optimization results under smoothness of the loss function and bounded variance of the stochastic gradients. Under these assumptions, the training dynamics satisfy:

- an $\mathcal{O}(1/T)$ convergence rate for full-batch gradient descent, and
- an $\mathcal{O}(1/\sqrt{T})$ convergence rate for mini-batch SGD, implying convergence to a first-order stationary point. Detailed derivations are provided in Appendix B.

V. SIMULATIONS

A. Training and Evaluation Setup

To evaluate our framework, we implemented and trained two distinct meta-learners, a XGBoost model, and a MLP model. The training and evaluation process followed the methodologies detailed in Section III-C.

a) Candidate STT Models: Our pool of candidate STT models consists of three SOTA systems: OpenAI's Whisper Large v3, NVIDIA's Parakeet-TDT 0.6B [45], and NVIDIA's Canary-Qwen-2.5B [46]. These models were selected as strong, general-purpose STT systems with different training objectives, architectures, and decoding pipelines, which naturally results in input-dependent performance variability across datasets and acoustic conditions. (e.g., see model cards and benchmark evaluations in [5], [45], [46]).

b) Baseline Combination Methods: To provide a comprehensive evaluation of ADASTT, we additionally compared our framework against standard and lightweight methods for combining multiple STT systems. In particular, we considered the following baselines, which do not rely on learned, input-dependent selection. (i) Random selection ensemble: For each input audio segment, a single STT model is selected uniformly at random from the candidate pool and used to generate the transcription. (ii) Recognizer Output Voting Error Reduction (ROVER) ensemble baseline: We implemented the ROVER algorithm [8], a widely used ensemble method for ASR system combination. ROVER aligns the word-level outputs of all candidate STT models and produces a consensus transcription via majority voting. Unlike ADASTT, ROVER requires running all STT models for every input and does not adapt its behavior based on input-specific acoustic characteristics.

Both baselines operate over the same pool of candidate STT models as ADASTT and are evaluated using identical preprocessing, normalization, and scoring pipelines. This ensures a fair comparison that isolates the effect of adaptive, feature-driven model selection from static or non-adaptive combination strategies.

c) Oracle Upper Bound Baseline: To quantify the theoretical headroom of routing-based selection, we also included an Oracle baseline that retrospectively selects, for each audio input, the STT model achieving the lowest WER. This provides an upper bound on performance achievable by any perfect routing mechanism over the given expert pool.

d) Training Data and Features: The meta-learners were trained on a composite dataset constructed from the training splits of four speech corpora: Common Voice 22.0 [47], VoxPopuli [48], LibriSpeech [9], and the AMI Meeting Corpus [49]. For VoxPopuli, we used the accented English subset and manually partitioned it into training, validation, and test splits with an 80:10:10 ratio, as no official training split is provided for this

variant. This collection was curated to expose the learners to a wide spectrum of acoustic conditions, ranging from clean speech (Common Voice, LibriSpeech) to noisy, multi-speaker, and accented scenarios (VoxPopuli, AMI). In total, our training set comprised 107,234 unique audio samples. For each sample, we extracted the comprehensive feature set described in Section III-B, including utterance-level signal statistics such as SNR computed using standard estimators implemented in widely used audio processing libraries.

e) MLP Meta-Learner: Our MLP meta-learner was designed as a deep neural network with five hidden layers, with dimensions decreasing from 1024 to 64. Each layer utilized a GELU activation function [43], followed by BatchNorm1d [50] and Dropout [51] for regularization, with dropout rates of 0.20, 0.10, 0.10, and 0.05 applied to the first four layers, respectively. The model was trained to minimize the expected WER objective defined in (3), using the AdamW optimizer [52] with an initial learning rate of $1e-4$ and weight decay of $1e-4$. Training was performed with a batch size of 1024, leveraging Automatic Mixed Precision (AMP) [53] for efficiency and gradient clipping with a max norm of 1.0 for stability. A ReduceLROnPlateau scheduler adjusted the learning rate based on the validation loss, and early stopping with a patience of 20 epochs was used to select the best-performing model checkpoint. All the hyperparameters are selected based on the validation dataset. The model was implemented in Pytorch [54].

f) XGBoost Meta-Learner: For comparison, the XGBoost meta-learner was implemented using a gradient tree-boosting procedure with the GPU-accelerated histogram algorithm for optimized performance. It was trained using the custom WER-based objective function derived in Section III-C. Key hyperparameters were set to a learning rate of 0.02, a maximum tree depth of 8, and a minimum child weight of 4. To enhance generalization and prevent overfitting, we employed a subsample ratio of 0.8, a column-sampling-by-tree ratio of 0.5, and an L2 regularization term of 1.0. Training also incorporated early stopping based on validation set performance. All the hyperparameters are selected based on the validation dataset.

g) Hardware: All experiments, including feature extraction, meta-learner training, and final evaluation, were conducted on a single L40S server hosted on the RunPod cloud platform.

B. Results on Test Sets

We evaluated ADASTT on the test splits of seven public datasets: LibriSpeech [9], AMI Meeting [49], Google FLEURS [55], Mozilla Common Voice (MCV) 22.0 [56], VoxPopuli [48], GigaSpeech [57], and Earnings 22 [58]. These datasets cover a wide range of acoustic conditions, speaker populations, and speech styles.

For each test, we compared ADASTT against three state-of-the-art STT models (Whisper Large v3, Parakeet TDT 0.6B, and Canary Qwen 2.5B) as well as two ensemble baselines, Random Ensemble selection and ROVER hypothesis fusion. To ensure strict evaluation consistency, all reported WERs were computed using a standardized text normalization pipeline, namely the Whisper's English normalizer, applied identically to all

baseline systems and ensemble outputs. The complete results are presented in Fig. 2 and Table I.

On the mixed dataset constructed by uniformly sampling from all evaluation corpora, ADASTT-XGB achieves the lowest WER of 0.1050, closely followed by ADASTT-MLP. Both methods outperform the best single model (Canary) as well as classical ensemble baselines (Random Ensemble, ROVER). Relative to the strongest ensemble baseline, ADASTT-XGB yields a 6.17% relative improvement. This advantage extends to Earnings22 and Mozilla Common Voice, where ADASTT surpasses the ROVER baseline while maintaining significantly higher computational efficiency by decoding only a single model per input. These results demonstrate that adaptive, per-input model selection provides a clear advantage over static ensemble strategies when heterogeneous acoustic conditions are jointly considered.

To better understand the behavior of the meta-learner across datasets, we also examined simple routing statistics. On several datasets a single backend already achieves the lowest WER for the vast majority of inputs, and the meta-learner correspondingly assigns most samples to that backend (e.g., over 94% of inputs on CommonVoice and about 95% on LibriSpeech-Clean are routed to the dominant model), with correspondingly low routing entropy. This behavior is consistent with the dataset structure and indicates that the selector reliably identifies the most suitable backend when a clear dominant model exists. In contrast, more heterogeneous datasets demonstrate more distributed routing decisions; for example, on AMI-SDM the dominant routing share drops to about 72%, while on the accented VoxPopuli split it further decreases to around 63% with routing entropy reaching 0.48. This reflects substantial expert diversity and highlights scenarios where adaptive model selection can provide the greatest benefit.

a) Robustness to Accent Shifts (VoxPopuli): The results on VoxPopuli (EN-Accented) highlight a critical failure mode of static model selection. While Canary Qwen 2.5B serves as the dominant expert on most benchmarks (e.g., AMI, Common Voice), it shows a significant degradation on accented speech, achieving a WER of 0.2129. A system statically deployed with Canary as the default “best” model would thus incur a substantial performance penalty in this domain.

In contrast, ADASTT successfully identifies this performance inversion. By dynamically routing samples to the superior expert (likely Parakeet in this regime), ADASTT-XGB achieves a WER of 0.1883, surpassing not only the struggling Canary model but also the best single expert (Parakeet) and the ROVER ensemble. This demonstrates ADASTT's ability to act as a safety mechanism, automatically shifting preference away from generally dominant models when they encounter specific acoustic conditions such as heavy accents where they lack robustness.

b) AMI Meeting (SDM): The AMI Meeting (SDM) dataset represents one of the most challenging evaluation conditions due to distant microphones, reverberation, and spontaneous multi-speaker interactions. Single-model performance varies substantially, with Canary outperforming Whisper by a wide margin. Ensemble methods offer limited benefits in this setting, as ROVER (0.2904) fails to improve over the Random Ensemble (0.2807), indicating that correlated errors dominate majority

TABLE I

WER (0–1) COMPARISON ACROSS DATASETS. LOWER IS BETTER. **BOLD** INDICATES THE BEST (LOWEST) WER AMONG {WHISPER, PARAKEET, CANARY, RANDENS, ROVER, ADASTT-XGB, ADASTT-MLP} FOR EACH ROW (ORACLE EXCLUDED). VALUES IN PARENTHESES (+%) FOR ADASTT INDICATE RELATIVE IMPROVEMENT OVER THE BEST COMPETING ENSEMBLE (MIN OF RANDOM ENSEMBLE AND ROVER) AS A PERCENTAGE, COMPUTED AS $(\frac{\text{BESTENS}-\text{ADASTT}}{\text{BESTENS}}) \times 100$. POSITIVE (GREEN) VALUES INDICATE IMPROVED PERFORMANCE. **ORACLE** DENOTES AN UPPER BOUND OBTAINED BY RETROSPECTIVELY SELECTING, FOR EACH INPUT, THE MODEL ACHIEVING THE LOWEST WER.

Dataset	Whisper	Parakeet	Canary	Random Ensemble	ROVER	ADASTT-XGB	ADASTT-MLP	Oracle
AMI Meeting (SDM)	0.3085	0.2971	0.2364	0.2807	0.2904	0.2509 (+10.62%)	0.2522 (+10.16%)	0.1491
Earnings22	0.1520	0.1409	0.1352	0.1427	0.1443	0.1340 (+6.10%)	0.1346 (+5.68%)	0.0953
Mozilla Common Voice	0.0719	0.0724	0.0575	0.0673	0.0634	0.0575 (+9.30%)	0.0581 (+8.36%)	0.0361
VoxPopuli (EN-Accented)	0.2144	0.1979	0.2129	0.2084	0.1900	0.1883 (+0.89%)	0.1895 (+0.26%)	0.1387
GigaSpeech	0.1142	0.1137	0.1151	0.1143	0.1155	0.1121 (+1.92%)	0.1119 (+2.10%)	0.0845
LibriSpeech Test-Other	0.0481	0.0422	0.0410	0.0438	0.0404	0.0413 (-2.23%)	0.0415 (-2.72%)	0.0247
LibriSpeech Test-Clean	0.0234	0.0207	0.0210	0.0217	0.0190	0.0211 (-11.05%)	0.0212 (-11.58%)	0.0123
Fleurs	0.0422	0.0417	0.0350	0.0397	0.0351	0.0350 (+0.28%)	0.0352 (-0.28%)	0.0240
Mixed Dataset	0.1218	0.1158	0.1068	0.1148	0.1122	0.1050 (+6.17%)	0.1055 (+5.72%)	0.0706
Average WER	0.1212	0.1154	0.1047	0.1138	0.1121	0.1042 (+7.04%)	0.1046 (+6.69%)	0.0691

voting. While ADASTT-XGB (0.2509) significantly outperforms these traditional ensembles (+10.62% relative improvement), it approaches the performance of the single dominant expert (Canary: 0.2364). This demonstrates that unlike hypothesis fusion, which is heavily penalized when weaker models outvote the dominant expert, ADASTT successfully identifies the superior expert, effectively mitigating the degradation typically observed in ensembles during high-variance regimes. As shown in Table I, the Oracle results further indicate that routing headroom exists in such high-variance conditions, particularly on AMI and accented VoxPopuli, where ADASTT captures a large portion of the attainable gain through adaptive expert selection.

c) *Low-Variance Regimes (GigaSpeech and Fleurs)*: On datasets where base model WERs are tightly clustered, such as GigaSpeech (0.1137–0.1155) and Fleurs (0.0350–0.0417), the potential headroom for selection-based gains is inherently limited. Consistent with this expectation, ADASTT yields only modest improvements (e.g., up to +2.10% on GigaSpeech) while matching the strongest baseline on Fleurs. Importantly, ADASTT never degrades performance in these low-variance regimes, confirming that the learned selection mechanism behaves conservatively when expert differentiation offers little benefit.

d) *LibriSpeech (Test-Clean and Test-Other)*: On LibriSpeech, recognition errors across models appear highly correlated, particularly on the Test-Clean split, as evidenced by the limited gains from hypothesis fusion. In this regime, ROVER achieves the lowest WERs. Nevertheless, ADASTT remains competitive and stable, demonstrating that adaptive selection does not introduce degradation even when the performance ceiling is largely saturated by conventional ensemble methods.

Averaged across datasets, ADASTT-XGB and ADASTT-MLP achieve the lowest mean WERs (0.1042 and 0.1046, respectively), surpassing all single-model and ensemble baselines. Relative to the strongest ensemble method, this corresponds to improvements of 7.04% and 6.69%. Overall, these results confirm that learned, adaptive selection provides a more robust and general strategy than static model selection or majority voting. Crucially, ADASTT addresses the inherent fragility of relying

on a single state-of-the-art model. While a highly optimized expert like Canary may excel on specific English benchmarks, static deployment creates a single point of failure when domain shifts occur (as observed in the accented VoxPopuli regime). Therefore, the primary contribution of ADASTT is not merely to rival a specific SOTA model, but to provide a model-agnostic meta-learning framework that is inherently extensible to any language or acoustic domain. Unlike static combination strategies such as Random Ensemble or ROVER, which lack input awareness and incur unnecessary computation, ADASTT is designed as an adaptive selection strategy that adapts to input-dependent variability by dynamically routing each input to the most suitable expert. This ensures consistent performance stability across high-variance regimes such as accented speech, and mixed evaluation sets where no single expert is uniformly optimal. These findings in high-variance regimes highlight the broader applicability of the proposed framework to domains with uneven expert quality, such as low-resource or multilingual ASR, without requiring any modification to the core selection architecture.

C. Algorithm Switching Analysis

To precisely evaluate the responsiveness and accuracy of the meta-learner’s selection mechanism, we designed a controlled experiment with two models. The goal of this experiment is not to assess generalization to unseen datasets, but to isolate and visualize the meta-learner’s ability to detect abrupt changes in input characteristics and to switch between candidate STT models without delay. These input characteristics are implicitly encoded through the extracted feature vectors, which jointly capture acoustic conditions such as background noise, speech style, temporal variability, and signal quality. To this end, We created a synthetic test sequence by concatenating audio clips where the superior model was known beforehand. We use Google Fleurs and GigaSpeech test splits to construct our setup. The sequence was structured in a repeating pattern: a block of approximately 50 audio clips on which Parakeet is known to perform better, followed by a block of 5 clips where Whisper Large v3 excels. This setup allows us to observe whether the meta-learners can

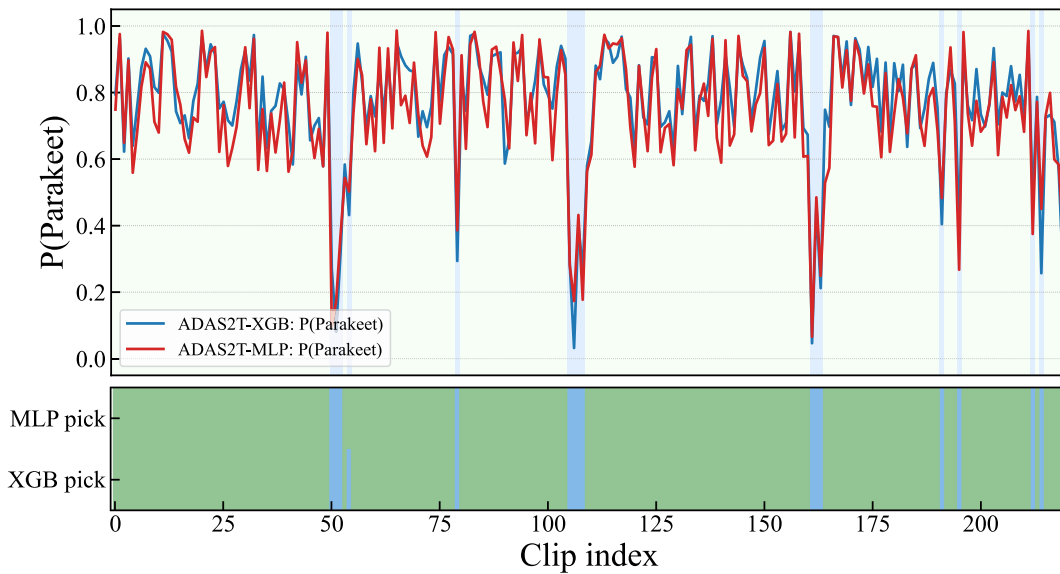


Fig. 3. Demonstration of ADASTT's switching behavior in a controlled experiment.

Top: Predicted probability of selecting Parakeet ('P(Parakeet)') by the XGBoost and MLP meta-learners. Both models show high probability for Parakeet-favored clips and a sharp drop for Whisper-favored clips (highlighted by light blue bands).

Bottom: The final model selected by each meta-learner. The picks (Green=Parakeet, Blue=Whisper) align precisely with the probability shifts, showing accurate, real-time adaptation.

detect the abrupt change in audio characteristics and switch their preferred model accordingly.

The results of this experiment are visualized in Fig. 3. The top panel displays the predicted probability of selecting the Parakeet model, $P(\text{Parakeet})$, generated by both the ADASTT-XGB (blue line) and ADASTT-MLP (red line) meta-learners. As shown, for the majority of clips, both models assign a high probability (on average 0.8) to Parakeet, correctly identifying the audio inputs where it has an advantage. These probability assignments arise from consistent shifts in the underlying feature distributions (e.g., higher noise levels, increased temporal variability, and conversational speech patterns) that were observed during training to favor Parakeet over Whisper. Crucially, at the boundaries of the inserted Whisper-favored clips (highlighted by light blue vertical bands), both meta-learners exhibit a sharp, immediate drop in $P(\text{Parakeet})$ to near zero. This demonstrates their high sensitivity to the features that differentiate the performance of the two STT models.

The bottom panel confirms this adaptive behavior by showing the final model selected by each meta-learner for every clip. The green bars indicate that Parakeet was chosen, while the blue bars signify a switch to Whisper. In the long segments, the system correctly selects Parakeet. As soon as a Whisper-favored clip is encountered, the system decisively switches its pick from Parakeet to Whisper, and then switches back just as quickly when the audio characteristics revert. This behavior demonstrates that ADASTT operates as a responsive, low-latency controller capable of real-time model routing, independently of the specific datasets used in this illustrative experiment.

D. Input-Level Routing Case Study

To improve the transparency of the routing mechanism, we provide a brief input-level case study illustrating how ADASTT

selects STT models for individual inputs. Importantly, the meta-learner does not use domain identifiers or dataset-specific rules; routing decisions depend solely on the extracted feature vector $\mathbf{x}_n$. Consequently, routing behavior can be interpreted by examining salient acoustic indicators (e.g., SNR, spectral entropy, silence ratio) together with the predicted routing probabilities.

Table II shows two representative input utterances drawn from acoustically contrasting regimes. For reproducibility, inputs were selected using a deterministic SNR-bin protocol defined prior to inspecting routing outcomes. For each input utterance we report interpretable acoustic indicators, the predicted routing distribution over models, the selected model, and the resulting per-model WERs.

Example A (Clean read speech). For a high-SNR input utterance from LibriSpeech Test-Clean, the acoustic indicators reflect relatively stable and low-entropy conditions. In this regime, the routing distribution becomes highly concentrated on a single model, and the selected model coincides with the best-performing backend. This behavior reflects the conservative routing strategy observed on low-variance datasets where base-model performance is tightly clustered.

Example B (Far-field conversational speech). For a challenging input utterance from the AMI-SDM meeting corpus, acoustic indicators show substantially lower SNR and higher entropy, reflecting reverberant far-field conditions. In this setting, model performance diverges more strongly. The routing distribution therefore shifts toward the model that empirically performs best in this regime, illustrating how ADASTT adapts its selection when acoustic variability increases.

These examples illustrate the core behavior of ADASTT: routing decisions remain conservative when model performance is similar, while probability mass shifts toward more robust models in acoustically challenging conditions. This qualitative behavior is consistent with the results reported in Table I, where

TABLE II

INPUT-LEVEL ROUTING CASE STUDY ILLUSTRATING ROUTING BEHAVIOR UNDER TWO CONTRASTING ACOUSTIC REGIMES. **SNR** DENOTES SIGNAL-TO-NOISE RATIO, **ENT**, SPECTRAL ENTROPY, AND **SIL**, THE SILENCE RATIO OF THE INPUT UTTERANCE. $\max_m p(m|\mathbf{x})$ INDICATES THE HIGHEST ROUTING PROBABILITY ASSIGNED BY THE META-LEARNER. **CHOSEN** DENOTES THE SELECTED STT MODEL. **WERS (W,P,C)** REPORT PER-MODEL WORD ERROR RATES FOR WHISPER (W), PARAKEET (P), AND CANARY (C), RESPECTIVELY. **ORACLE MATCH** INDICATES WHETHER THE SELECTED MODEL COINCIDES WITH THE LOWEST-WER EXPERT FOR THAT INPUT UTTERANCE.

Regime	SNR	Ent.	Sil.	$\max_m p(m \mathbf{x})$	Chosen	WERS (W,P,C)	Oracle Match
A (Clean)	16.629	3.719	0.400	0.9728 (C)	C	(0.400, 0.400, 0.000)	Yes
B (Far-field)	1.982	3.942	0.827	0.6794 (P)	P	(0.833, 0.583, 1.000)	Yes

TABLE III

AVERAGE INFERENCE LATENCY (SECONDS PER SAMPLE). COMPARISON OF BASE STT MODELS, ENSEMBLE BASELINES, AND ADASTT ROUTING VARIANTS. **FEATURE** REPRESENTS THE DATASET-SPECIFIC FEATURE EXTRACTION COST, WHILE THE REPORTED ADASTT INFERENCE TIMES INCLUDE FEATURE EXTRACTION, THE META-LEARNER SELECTION OVERHEAD (≈ 2 MS) AND BASE MODEL INFERENCE. THE MIXED DATASET IS FORMED BY COMBINING EQUAL-SIZED SAMPLES FROM THE EVALUATED DATASETS IN THIS EXPERIMENT.

Dataset	Feature	Parakeet	Whisper	Canary	RandEns	ROVER	ADASTT-XGB	ADASTT-MLP
AMI Meeting-SDM	0.0651	0.0435	0.2971	0.1511	0.1639	0.5117	0.1922	0.1882
LibriSpeech-Other	0.1249	0.0491	0.5803	0.4440	0.3578	1.0934	0.5659	0.5504
LibriSpeech-Clean	0.1845	0.0502	0.6331	0.5000	0.3944	1.2033	0.6853	0.6797
Fleurs	0.2483	0.0520	0.6873	0.5200	0.4198	1.2793	0.7723	0.7733
Mozilla Common Voice	0.1441	0.0465	0.3682	0.2206	0.2118	0.6553	0.5363	0.3614
VoxPopuli	0.2737	0.0530	0.8019	0.5770	0.4773	1.4519	0.8316	0.8145
GigaSpeech	0.1416	0.0468	0.8625	0.3773	0.4289	1.3066	0.5264	0.5491
Mixed	0.1468	0.0452	0.7230	0.3989	0.3890	1.1871	0.5368	0.5311

the largest gains occur on heterogeneous datasets such as AMI and accented VoxPopuli.

E. Inference Time Analysis

In order to quantify the computational efficiency of the ADASTT framework at inference time, we measured the average end-to-end inference time per audio input across all evaluation datasets, decomposing the total into three components: feature extraction, meta-learner model selection, and execution of the chosen STT model. The purpose of this analysis is to quantify the overhead of adaptive selection rather than to claim uniform speedups over all baselines. While some datasets favor a faster model, others exhibit mixed routing behavior, and in all cases ADASTT avoids the multiplicative latency of running multiple STT models per input. The detailed total inference times for all datasets are reported in Table III. The meta-learner selection time is included in the reported ADASTT inference times and is not listed separately due to its negligible cost (< 2 ms per input).

Across all datasets, feature extraction incurs between 0.065 s and 0.274 s per audio input. As a result, total inference time is dominated by the execution cost of the selected base STT model. On the mixed evaluation set, the complete ADASTT-MLP pipeline, consisting of feature extraction, selection, and single model inference, is approximately 26.5% faster than running Whisper Large V3 alone, while simultaneously achieving lower WER. Compared to the ROVER baseline at 1.19 s, ADASTT-MLP at 0.53 s yields a 55.3% reduction in end to end latency. Importantly, on the AMI Meeting SDM corpus, ADASTT MLP is nearly $2.7\times$ faster than the ROVER ensemble.

More broadly, adaptive selection enables a flexible efficiency trade-off compared to static deployments. While lightweight models remain the fastest option for simple tasks, ADASTT

significantly reduces latency compared to computationally intensive SOTA models (e.g., Whisper) by identifying instances where lightweight experts outperform larger architectures, thereby avoiding the computational cost of larger models when they provide no accuracy benefit. This stands in sharp contrast to hypothesis fusion methods such as ROVER, which requires running all constituent models and incur additive latency. By running only one expert per input, ADASTT eliminates the cumulative delay of traditional ensembles, making it fast enough for real-time applications.

F. Ablation Study

We conducted an ablation analysis to quantify the contribution of each feature group introduced in Section III-B: spectral-phonetic features (MFCCs), prosodic features, neural speech embeddings, and signal-temporal complexity features. The baseline configuration includes all four groups jointly. The complete set of ablation results is provided in Table IV.

The XGBoost meta-learner demonstrates a strong dependency on high-level representations. Removing Neural Embeddings (NE) results in the most significant degradation for XGB. This impact is particularly visible on acoustically diverse datasets like VoxPopuli and AMI. This behavior indicates that tree-based meta-learners benefit substantially from high-level semantic and speaker-related information encoded in pretrained neural speech embeddings when operating under challenging acoustic conditions.

In contrast, the MLP-based router relies more heavily on low-level spectral cues, with the largest performance degradation observed when MFCCs or signal temporal features are removed. Interestingly, excluding neural embeddings leads to a slight improvement in the MLP's average performance. This

TABLE IV

ABLATION STUDY RESULTS FOR MLP AND XGB MODELS ACROSS VARIOUS DATASETS. THE VALUES REPRESENT THE WORD ERROR RATE (WER) (LOWER IS BETTER). THE ROWS SHOW THE PERFORMANCE WHEN SPECIFIC FEATURE GROUPS ARE REMOVED (ABLATED). **ABBREVIATIONS: FEATURES:** MFCC (MEL-FREQUENCY CEPSTRAL COEFFICIENTS), NE (NEURAL EMBEDDINGS), PV (PROSODIC & VAD FEATURES), ST (SIGNAL TEMPORAL FEATURES). **DATASETS:** AMI (AMI MEETING), LS-OTH (LIBRISPEECH TEST-OTHER), LS-CLN (LIBRISPEECH TEST-CLEAN), MCV (MOZILLA COMMON VOICE), VoxPop (VOXPOPULI), GIGA (GIGASPEECH).

Ablated Features	Model	Dataset									
		AMI	LS-C	LS-O	MCV	Fleurs	VoxPop	Giga	Earnings22	Mixed	Average
Full Model (Baseline)	MLP	0.2522	0.0212	0.0415	0.0581	0.0352	0.1895	0.1119	0.1346	0.1055	0.1046
	XGB	0.2509	0.0211	0.0413	0.0575	0.0350	0.1883	0.1121	0.1340	0.1050	0.1042
<i>Single Feature Ablations</i>											
— ST	MLP	0.2534	0.0214	0.0411	0.0580	0.0353	0.1894	0.1128	0.1369	0.1060	0.1052
	XGB	0.2530	0.0211	0.0413	0.0575	0.0350	0.1882	0.1122	0.1329	0.1051	0.1043
— MFCC	MLP	0.2558	0.0210	0.0407	0.0577	0.0353	0.1896	0.1124	0.1390	0.1064	0.1057
	XGB	0.2538	0.0211	0.0411	0.0580	0.0350	0.1831	0.1131	0.1339	0.1049	0.1045
— PV	MLP	0.2549	0.0211	0.0409	0.0579	0.0351	0.1882	0.1118	0.1344	0.1055	0.1048
	XGB	0.2515	0.0211	0.0413	0.0575	0.0350	0.1868	0.1117	0.1330	0.1047	0.1040
— NE	MLP	0.2492	0.0207	0.0407	0.0574	0.0350	0.1872	0.1090	0.1370	0.1045	0.1038
	XGB	0.2540	0.0215	0.0412	0.0575	0.0350	0.1948	0.1115	0.1354	0.1064	0.1051

behavior may indicate sensitivity to high-dimensional embedding features under simple feature concatenation, potentially leading to reduced generalization.

Overall, the results demonstrate that the feature groups contribute complementary and context-dependent information to the meta-learner rather than exhibiting a uniform importance ranking across datasets. The combined use of heterogeneous acoustic, prosodic, and embedding-based features enables ADASTT to adapt its model-selection behavior to varying acoustic conditions, and removing any single group degrades performance in at least a subset of evaluation settings.

VI. CONCLUSION

We studied the problem of robust real-time STT transcription under diverse and unpredictable acoustic conditions. As different STT models show variable performance across input types, depending on factors such as noise level, speaker characteristics, and recording quality, no single model uniformly outperforms others across all scenarios. Moreover, exhaustively applying all candidate models to each audio input is computationally expensive in real-time systems. To this end, we introduced ADASTT, a meta-learning-based model selection framework that dynamically selects the most suitable STT model for each audio input based on input-specific features. These features, which include low-level acoustic descriptors, prosodic indicators, and contextual embeddings, are extracted with minimal computational overhead. ADASTT employs a probabilistic meta-learner to predict which transcription model is likely to maximize the transcription accuracy for the given input. Although our framework is learner-agnostic, we presented two realizations of the meta-learner: (i) an XGBoost based meta-learner trained with a custom WER-based objective and (ii) a neural network

based meta-learner. We provided convergence guarantees for both training procedures and showed that feature extraction adds only small computational complexity over a single STT forward pass. We evaluated ADASTT across multiple noisy and heterogeneous datasets, where it demonstrated superior robustness compared to static model selection and traditional ensembles, achieving the lowest WER on the aggregated mixed evaluation set. Furthermore, by avoiding the computational redundancy of hypothesis fusion, ADASTT maintains a lightweight inference profile, making it a viable and efficient strategy for latency-sensitive applications.

APPENDIX A

CONVERGENCE OF THE XGBOOST META-LEARNER

In this appendix, we provide the detailed convergence analysis for the XGBoost-based meta-learner.

a) *Preconditions (A1–A3):*

A1 *Bounded cost:* Recall that each of K candidate STT algorithm M_m produces a transcription $T_{n,m}$ for input a_n , and the associated per-model cost is defined as

$$c_{n,m} := \text{WER}(T_{n,m}, R_n),$$

the Word Error Rate (WER) relative to reference R_n , where the WER is defined as in Section III-C. We assume two uniform bounds hold:

$$0 \leq c_{n,m} \leq B_{\max} := \max_{n,m} c_{n,m},$$

$$|c_{n,i} - c_{n,j}| \leq B_{\text{rng}} := \max_n \max_{i,j} |c_{n,i} - c_{n,j}|.$$

The second derivative term is

$$h_{n,m} = w_{n,m}(1 - 2w_{n,m})(c_{n,m} - L_n),$$

where $L_n = \sum_j w_{n,j} c_{n,j}$ is a convex combination of the costs. Thus

$$c_{n,m} - L_n = \sum_{j \neq m} w_{n,j} (c_{n,m} - c_{n,j}),$$

$$|c_{n,m} - L_n| \leq (1 - w_{n,m}) B_{\text{rng}}.$$

Therefore,

$$\begin{aligned} |h_{n,m}| &\leq B_{\text{rng}} w_{n,m} (1 - w_{n,m}) |1 - 2w_{n,m}| \\ &\leq \frac{\sqrt{3}}{18} B_{\text{rng}} \\ &< \frac{B_{\text{rng}}}{8}. \end{aligned}$$

For simplicity, we state the clean bound

$$|h_{n,m}| \leq \frac{B_{\text{rng}}}{8}.$$

After clipping $h_{n,m}$ below by $\varepsilon > 0$, all diagonal entries of the approximate Hessian H^* lie in $[\varepsilon, B_{\text{rng}}/8]$, giving

$$\varepsilon I \preceq H^* \preceq \frac{B_{\text{rng}}}{8} I.$$

Finally, we assume that the overall loss L is L_g -smooth, i.e., its gradient is Lipschitz with constant $L_g > 0$.

A2 *Weak-learner edge*: Let $z^{(t)} \in \mathbb{R}^{N \times K}$ be the prediction matrix at boosting iteration t , and write

$$g_t := \nabla_z L(z) \big|_{z=z^{(t)}} \in \mathbb{R}^{N \times K}.$$

Let f_t be the depth ≥ 1 regression tree obtained by least-squares fitting to the pseudo-targets $-g_t$, and update

$$z^{(t+1)} = z^{(t)} + \eta f_t, \quad \eta > 0.$$

We assume that f_t satisfies the *edge condition*

$$\langle -g_t, f_t \rangle \geq \rho \|g_t\|_2^2,$$

Here, $\|\cdot\|_2$ denotes the empirical ℓ_2 norm over (n, m) pairs, and the constant $\rho > 0$ depends on the tree depth and the chosen regularisation (see [20]). In particular, with row/column subsampling, the inequalities are applied to the subsampled objective. In practice the effective edge ρ depends on the sampling rate, tree depth, and regularization.

A3 *Quadratic fit bound*: The regularized least-squares fit in XGBoost satisfies

$$\|f_t\|_2^2 \leq \lambda^{-1} \|g_t\|_2^2,$$

where $\lambda > 0$ is the ℓ_2 regularization coefficient in the leaf-weight formula [41].

b) *Descent Lemma*: Let $L_t := L(z^{(t)})$ denote the training loss evaluated at the current logits $z^{(t)}$. By the smoothness assumption from A1, the gradient $\nabla_z L$ is L_g -Lipschitz; equivalently, for any $v \in \mathbb{R}^{N \times K}$ and some ξ on the segment $[z^{(t)}, z^{(t+1)}]$,

$$v^\top H_z(\xi) v \leq L_g \|v\|_2^2.$$

Thus, using a second-order Taylor expansion and the smoothness assumption:

$$L_{t+1} \leq L_t - \eta \rho \|g_t\|_2^2 + \frac{1}{2} \eta^2 L_g \|f_t\|_2^2.$$

Substituting A3:

$$L_{t+1} \leq L_t - \eta \left(\rho - \frac{\eta L_g}{2\lambda} \right) \|g_t\|_2^2. \quad (7)$$

Define

$$\gamma := \eta \rho - \frac{\eta^2 L_g}{2\lambda}.$$

Then, choosing $\eta < \frac{2\rho\lambda}{L_g}$ ensures $\gamma > 0$ and hence:

$$L_{t+1} \leq L_t - \gamma \|g_t\|_2^2.$$

c) *Convergence Rate*: Summing (7) over t gives:

$$\sum_{t=0}^{T-1} \|g_t\|_2^2 \leq \frac{L_0}{\gamma} \Rightarrow \min_{0 \leq t < T} \|g_t\|_2^2 \leq \frac{L_0}{\gamma T},$$

ensuring a convergence rate of $O(1/T)$ toward a first-order stationary point. Since L is non-convex, this does not imply global optimality.

APPENDIX B

CONVERGENCE OF THE MLP META-LEARNER

In this appendix, we provide the convergence analysis for the neural (MLP-based) meta-learner.

a) *Preconditions (A1–A3)*: Recall that each of K candidate STT algorithm M_m produces a transcription $T_{n,m}$ for input a_n , and the associated per-model cost is defined as

$$c_{n,m} := \text{WER}(T_{n,m}, R_n),$$

the Word Error Rate (WER) relative to reference R_n , where the WER is defined as in Section III-C. Let $L(\theta)$ be defined as the total loss over the training set for the meta-learner with parameters θ :

A1 *Bounded loss*:

$$\begin{aligned} 0 &\leq L(\theta) \leq B := \max_{n,m} c_{n,m} \\ &= \max_{n,m} \text{WER}(T_{n,m}, R_n). \end{aligned}$$

A2 *L_g -smooth gradient*: Assume ∇L is L_g -Lipschitz, i.e.,

$$\|\nabla L(\theta) - \nabla L(\theta')\| \leq L_g \|\theta - \theta'\| \quad \text{for all } \theta, \theta'.$$

A3 *Bounded stochastic-gradient variance*:

$$\mathbb{E} \|g_t - \nabla L(\theta_t)\|^2 \leq \sigma^2,$$

for some $\sigma^2 \geq 0$, i.e., the stochastic gradients are unbiased with bounded variance.

Let $L^* := \inf_\theta L(\theta)$. Under A1–A3, the standard assumptions required in the non-convex analyses of Bottou–Curtis–Nocedal [59] and Ghadimi–Lan [60] are satisfied.

Remark 6: Our analysis assumes $L(\theta)$ is L_g -smooth, which holds when using smooth activations such as GELU or Softplus.

In practice, we train with AdamW [52] for efficiency, though the convergence proof is stated for (stochastic) gradient descent.

b) One-step Descent Lemma: For the full-batch update $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$ with any fixed step size $0 < \eta \leq 1/L_g$, the loss decreases at every step according to:

$$L(\theta_{t+1}) \leq L(\theta_t) - \left(\eta - \frac{L_g}{2}\eta^2\right) \|\nabla L(\theta_t)\|^2. \quad (8)$$

c) Full-batch Convergence: Summing (8) for $t = 0, \dots, T-1$ gives

$$\min_{0 \leq t < T} \|\nabla L(\theta_t)\|^2 \leq \frac{L(\theta_0) - L^*}{T\eta(1 - \frac{1}{2}\eta L_g)} = \mathcal{O}(T^{-1}). \quad (9)$$

d) Mini-batch SGD Convergence: With $\eta_t = \eta_0/\sqrt{t+1}$, $\eta_0 \leq 1/L_g$, and using A3,

$$\begin{aligned} \mathbb{E} \left[\min_{0 \leq t < T} \|\nabla L(\theta_t)\|^2 \right] &\leq \frac{2(L(\theta_0) - L^*) + \eta_0 L_g \sigma^2 \log T}{\eta_0 \sqrt{T}} \\ &= \mathcal{O}(T^{-1/2}), \end{aligned} \quad (10)$$

where σ^2 is the stochastic-gradient variance bound specified in A3. Because L is non-convex, (9)–(10) guarantee convergence only to a first-order stationary point, which is sufficient for training neural meta-learners [60].

In summary, under standard smoothness and boundedness assumptions, both the XGBoost-based and MLP-based meta-learners enjoy convergence to first-order stationary points of their respective training objectives, providing theoretical support for the stability of the proposed ADASTT meta-selection framework.

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